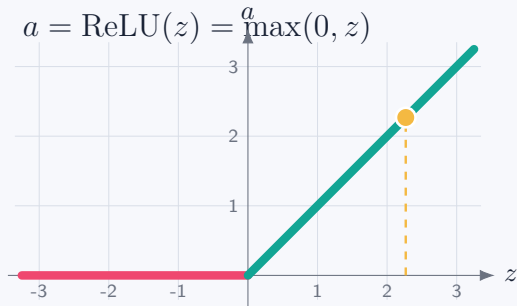


ReLU controls signal and gradient flow

FORWARD PASS

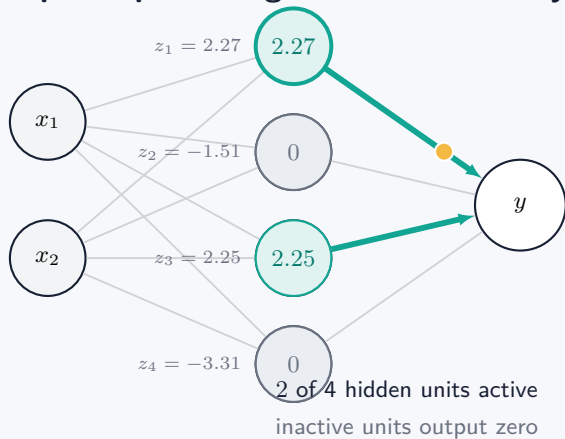
Input-dependent gates in a hidden layer



$$\text{ReLU}(z) = \begin{cases} 0, & z \leq 0 \\ z, & z > 0 \end{cases}$$

$$\text{ReLU}'(z) = \begin{cases} 0, & z < 0 \\ 1, & z > 0 \end{cases}$$

$$z = \mathbf{2.27} \quad a = \mathbf{2.27} \quad \text{ReLU}'(z) = \mathbf{1}$$



$$z = \mathbf{w}^\top \mathbf{x} + b \quad \mapsto \quad a = \max(0, z) = \mathbf{2.27} \quad \mapsto \quad y$$